

addition, WD has been widely employed in the context of reliability physics and engineering [30]. Its use in the context of the statistical analysis of RRAM makes sense since it is a weakest-link type distribution, i.e., the failure of the whole is dominated by the degradation rate for the weakest element.

The cumulative distribution function for the WD is given in Equation 1 [8, 30]

$$F(\nu) = 1 - \exp\left(-\left(\frac{\nu}{\alpha}\right)^\beta\right) \quad (1)$$

For the devices reported above the statistical analysis based on the WD has been performed. On the one hand, the V_{reset} and I_{reset} for all the reset curves of the 2749 cycles under consideration in our RS series were computed [8, 31]. On the other hand, the typical Weibits, calculated as $\ln[-\ln(1 - F)]$, have been obtained. If Weibits are plotted versus $\ln(V_{\text{reset}})$, a linear plot should be obtained if experimental data follow a WD, where the slope corresponds to the β parameter in Equation 1 (β measures the statistical dispersion [8, 30]). The results obtained for our devices are shown in Figure 2.

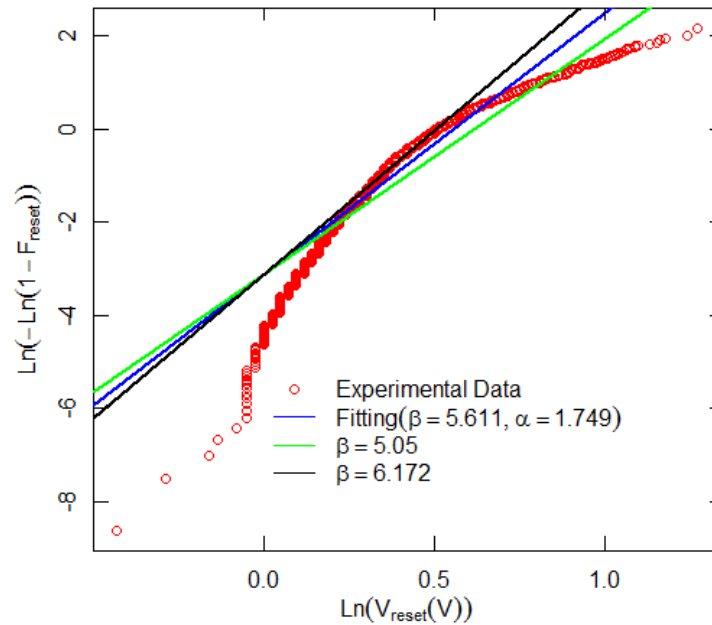


Figure 2. WD linear fit of V_{reset} for 2749 RS cycles. The best fit is obtained with the blue line, a 10% reduction (increase) in the beta parameter was assumed in the green (black) line.

Other analytical distributions (Equation 1) were also included making use of a β parameter with a 10% variation with respect to the best fit obtained in the statistical analysis. As can be seen, the Weibits of the experimental data are not linear. Therefore, although a rough approximation could be performed in the WD context, it seems reasonable to try other distributions. We do so in the following section and we call the reader's attention to the fact that a much better fit can be obtained with a Phase-type distribution modelling. After a progressive analysis based on step-by-step estimation of phase-type distributions [22, 23], it will be shown that the ED provides the best fit. Other previous statistical analysis on RRAMs might work much better making use of the ED; however, in this manuscript the study is limited to our experimental data.

3. THEORY AND METHODS

As shown in previous section, the logarithm of the experimental cumulative hazard rate versus $\ln(V_{\text{reset}})$ is not linear and therefore Weibull distribution seems not to be the appropriate distribution for fitting to V_{reset} . Then, the aim of this section is to find out what is the best probability distribution that describes the RRAM variability.

3.1. Phase-type distributions

One class of non-negative probability distributions with very interesting properties that allows to describe the main associated measures in an algorithmic form and to interpret the results is the phase-type distribution class. Phase type distributions were introduced and described in detail by [22] and [23].

The flexibility of the phase-type distribution makes it a good candidate to try a better fit since it generalizes a great number of distributions. The usefulness of this distribution class has been proved in several fields such as queueing theory, renewal processes,

reliability and survival [32, 33, 34, 35, 36]. In our case, we can assume that for our devices the conductive filament within the RRAM dielectric pass through different degradation stages before the rupture process takes place (absorption, in the approach we are following). At this point it seems reasonable to figure out the evolution of the conductive filament, i.e., the different stages followed in the destruction process versus V_{reset} .

A phase type distribution (PHD) is defined as the distribution of the lifetime up to the absorption in an absorbing Markov process (voltage up to the conductive filament failure in the RRAM context).

In the context of RRAMs an absorbing Markov process to model the voltage to the failure of the conductive filament can be assumed. The state space is given by a general number of m transient degradation stages, where the probability of being initially in stage i is given by α_i and one absorbing state, $m+1$, which is the conductive filament failure. In addition, the transition intensity from the transient stage i to the transient stage j is given by q_{ij} for $i \neq j$ and if $i=j$ then $q_{ii} = -\sum_{\substack{j=1 \\ j \neq i}}^{m+1} q_{ij}$. The voltage up to failure is

PHD distributed with representation $(\mathbf{a}, \mathbf{T})$ being $\mathbf{a} = (\alpha_1, \dots, \alpha_m)$; $\mathbf{T} = (q_{ij})_{i,j=1,\dots,m}$.

A PHD is a non-negative probability distribution whose cumulative distribution function is given in Equation 2

$$F(v) = 1 - \mathbf{a} \exp(\mathbf{T}v) \mathbf{e}, \quad v \geq 0, \quad (2)$$

where $\mathbf{e}$ is a column vector of ones with appropriate order. The density function associated to this distribution is

$$f(v) = -\mathbf{a} \exp(\mathbf{T}v) \mathbf{T} \mathbf{e} = \mathbf{a} \exp(\mathbf{T}v) \mathbf{T}^0, \quad v \geq 0,$$